

WASHINGTON UNIVERSITY IN ST. LOUIS
OLIN BUSINESS SCHOOL

DAT 5564: Database Design and SQL
Fall 2025, Mini A
In class Lab #1: E/R Diagrams

INSTRUCTIONS

1. This is a group assignment to be completed during the lab.
2. Each member needs to answer the questions on Canvas.
3. Submission for this lab will be digital.
 - a. The submission is due by the lab, on Canvas.

SESSION OUTLINE

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| • Introduction | 15 minutes |
| • Selecting groups | 5 minutes |
| • Lecture: Short review and Subclasses | 15 minutes |
| • Assignment Questions | 30 minutes |
| • Discuss solution approaches | 15 minutes |

ASSIGNMENTS

Submit answers on Canvas according to the description.

We are responsible for designing a database for facilities for the summer Olympics, as well as the events in those facilities. Include several attributes describing each facility. These facilities are called “sports complexes”. A sports complex may be either for a single sport or multiple sports. For example, the basketball facility has only a single sport. Multisport complexes have areas of the complex designated for each sport with a location indicator (e.g., middle, South-East corner, etc.). Include in the list of attributes of the sports complex the following attributes: name, GPS location, manager, maximum occupancy, **and add 2 more attributes**.

Each sports complex holds multiple events related to one or more sports. For example, the gymnasium has men’s floor exercise, women’s horizontal bars, women’s parallel bars, etc. Each sport has several attributes, including: name, gender of participants, **and add 2 more attributes**. When a sporting event is scheduled in a facility, the date and time of the event must be included.

Part 1: Design the corresponding E/ER diagram. Remember throughout to include arrows representing cardinality, where relevant. Underline key attributes. Also, convert the diagram to relations. In other words, show the tables that would be used in the database.

Part 2: Modify the previous E/ER diagram to include the referees for each sport. Assume that several referees are dedicated to a sport. In other words, a referee is only qualified to evaluate one sport. For now, the only attribute we can about is the referee's ID. This ID is a unique identifier for the referee. However, your diagram should clarify the sport the referee is assigned to.

Part 3: Modify your previous E/ER diagram to include additional attributes for referees. List at least 2 attributes for each referee.